

Research paper

Influences of geometry configurations on the performance of micro-nozzles

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ABSTRACT

Micro-nozzles play a critical role in various applications, such as micro-/nano-satellites in aerospace and heat dissipation and cooling in microelectronic systems (MEMS). The flows inside micro-nozzles can experience both continuum and rarefied flow due to their small size. In this research, we explore the optimal geometry configuration of micro-nozzles to achieve comprehensive performance and size, as well as background pressure compensation. Using classical computational fluid dynamics (CFD) and direct simulation Monte Carlo methods (DSMC), we investigate the effects of the expansion ratio, cross-profile shape, and plug on the performance of micro-nozzles operating over a wide range of pressures (0.5 – 100 kPa). The results reveal that micro-nozzle performance is influenced by the interplay between flow expansion and viscous loss induced by the subsonic layer next to the micro-nozzle wall. The findings show that micro-nozzle performance improves with increasing expansion ratio in slip and continuum flow regimes, but decreases in the transition flow regime. Additionally, the results indicate that the axisymmetric micro-nozzle (with a circular cross-profile) outperforms the linear micro-nozzle (with a rectangular cross-profile) in terms of both performance and size when subject to the same conditions, making it the recommended choice for practical applications. We also find that plugs designed at the throat decrease performance, but increase stability under diverse background pressures. This study provides valuable insights for the design of high-efficiency micro-nozzles in applications such as nanosats and MEMS heat dissipation units.

1. Introduction

Micro-nozzles are extensively used in aerospace, as well as flow control [1] and heat dissipation units for microelectronic systems (MEMS) [2]. In aerospace, micro-/nano-satellites have significant advantages over traditional large-scale satellites due to their small size, low power consumption, short development cycle, satellite formation capabilities, and ability to accomplish complex space missions at a lower, making them suitable for scientific research, defense, and commercial applications [3]. A De Laval nozzle accelerates the compressible gas to supersonic speeds in the axial direction by converting the thermal energy of the flow into kinetic energy, generating a force (thrust) [4]. Micro-nozzles, scaled down from traditional macroscopic De Laval nozzles to the micrometer scale, are a crucial component of the micro-propulsion system of micro-/nano-satellites.

Micro-nozzles, with a typical throat diameter of 10^{-4} – 10^{-5} m, are subject to rarefied effects resulting from their reduced size, which differs from traditional macroscopic nozzles, making the work characteristics, including the thrust, specific impulse, and efficiency, highly dependent on their geometric features [5]. Many experimental investigations of the micro-nozzles have been accomplished across a wide

range of Reynolds numbers (Re) [6–10]. Here, $Re = \rho v d_t / \mu_t$ characterizes the gas inertia to viscous effect at the nozzle throat, where d_t refers to the throat diameter; ρ , v , and μ_t stand for the density, velocity, and dynamic viscosity of the gas. Ketsdever et al. [7] experimentally demonstrated that the influence of the micro-nozzle diverging length and expansion half-angle is negligible when the Re is 200. However, as Re is less than 60, reducing the length of the nozzle enhances the performance of the micro-nozzle. Brucoleri et al. [8] investigated the thrust efficiency of micro-nozzles, examining various lengths and depths of the expansion section, as well as conical and bell-shaped configurations, within a Re range of 15 to 2000. All of the micro-nozzles show that the thrust efficiency remains around 75% for Re above 1500, but it drops to about 60% at a Re of 400. The efficiency continues to drop to below 40% when Re falls below 250. The authors observed that there are no substantial differences in performance between conical and bell-shaped micro-nozzles. Nishii et al. [9] revealed that the discharge coefficient and the Re had a positive correlation for a conical micro-nozzle working at $34 \leq Re \leq 763$. Furthermore, the authors highlighted that the decrease in the operating and background pressure ratio is an important factor in the decrease in the specific impulse of the

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micro-nozzle. Nishii et al. [10] also measured the performance of a micro-nozzle using water vapor as the propellant with the Re ranging from 100 to 600. The findings reveal that the discharge coefficient for the micro-nozzle using water vapor as the propellant remains approximately constant at 1. This differs from the micro-nozzles employing argon and nitrogen propellants and is attributed to an increased mass flux resulting from the two-phase flow of water within the nozzle.

The influence of objective factors such as vibrations [6], vacuum system [9], and gauge accuracy [8] can lead to relatively large experimental errors, making it difficult to draw quantitative conclusions for micro-nozzles with Re below 1000. Hence, many researchers have conducted numerical simulation studies on micro-nozzles covering a wide range of Re . Louissos et al. [11–14] conducted an extensive numerical investigation of the linear micro-nozzle developed at the Goddard Space Flight Center (GSFC) of the National Aeronautics and Space Administration (NASA). They investigated the effects of expander half-angles on the micro-nozzle performance at a background pressure of 1 kPa and operating pressures ranging from 5 kPa to 250 kPa. They found that tradeoffs exist between the subsonic boundary generated by gas viscosity and the flow deviation generated by expander half-angles, resulting in maximum thrust at an expander half-angle of approximately 30 degrees [11,12]. The influence of micro-nozzle depths (25–400 μm) was also explored. The results show that thrust and specific impulse efficiency increase with an increase in micro-nozzle depth. Three different bell expander configurations were examined, showing that the 30 deg linear micro-nozzle outperforms the bell micro-nozzle for most flow scenarios examined (i.e., nozzle depth and Re) [13]. San et al. [15] investigated micro-nozzles with different geometry sizes and expansion ratios, achieved by varying throat sizes and expansion half-angles. The analysis demonstrates that flow velocity in the diverging section of the micro-nozzle decreases as the geometry size decreases, and the sonic region moves away from the throat to the outlet because of the higher viscous dissipation in micro micro-nozzle. The Mach number (Ma) does not alter much at the micro-nozzle exit with various expansion ratios at the operating pressure of 25 kPa. The findings suggest that reducing the viscosity in the micro-nozzle can effectively enhance the performance of the micro-nozzle. Specifically, it can be achieved through the following means: the heat exchange between the gas and the surrounding substrate [5], increasing the Re [16,17], expanding the throat diameter [5,17], ensuring a smooth inner wall [18], and raising the inlet-to-outlet pressure ratio [19]. However, the specific relationship between the expansion ratio and micro-nozzle performance is not revealed. In addition, the expansion ratio can be changed by varying the diverging length, which is a crucial parameter for controlling the device size.

To our knowledge, current research [20] focuses mainly on linear micro-nozzles, which can be fabricated using common micro-machining techniques such as reactive ion etching and wet etching. These techniques enable the planar processing of various fine structures [21,22]. The cross-profile shape of two-dimensional (2-D) linear micro-nozzles differs from that of traditional large-scale axisymmetric De Laval nozzles. While axisymmetric nozzles have an advantage over linear nozzles in terms of size at a constant expansion ratio, the aforementioned techniques are difficult to apply to axisymmetric processing. However, some convenient and highly accurate methods [23,24] for fabricating axisymmetric micro-nozzle are now available, such as excimer laser micro-machining [25]. Despite this, there are still deficiencies and limitations in the performance study of axisymmetric micro-nozzles and their potential for development in the field of micro-/nano-satellites and micro-heat dissipation.

The micro-nozzle cooling system, which benefits from a high local cooling rate and better temperature uniformity compared to micro-channel coolers, is an effective cooling technique for high-power electronics [26]. Under certain conditions, such as in silicon processing and mechanical micro-machining, operations are conducted at low-background pressure environments. A critical challenge in the application of high-precision heat sinks is ensuring consistent spray cooling

performance from micro-nozzles. The functioning of gas-based cooling devices can inadvertently alter the ambient conditions (due to the expulsion of gas) by increasing the background pressure, which necessitates the reliable operation of micro-nozzles under the fluctuating background pressures brought about by ongoing spray cooling. Aerospike and plug nozzles are examples of rocket nozzles that have been considered for single-stage-to-orbit systems [27,28]. These types of nozzles are superior in their ability to compensate for background pressure. Micro-nozzles with a plug could be used as a developable and effective radiator component because of their stable performance at low-pressure ratios (i.e., the ratio of operating pressure to background pressure). However, previous studies have mainly focused on a given background pressure, and there is still a lack of research on the operating performance of micro-nozzles under different background pressure conditions.

In this study, we numerically investigate the effects of the expansion ratio, cross-profile shape, and plug on the thrust and specific impulse of micro-nozzles, operating at a wide range of operating pressures (0.5–100 kPa). We use the classical computational fluid dynamics (CFD) method for micro-nozzles that undergo continuum and slip flow and the direct simulation Monte Carlo (DSMC) method for transition flow. Our research on expansion ratio and cross-profile shape aims to identify geometrical configurations that balance micro-nozzle performance and size. Moreover, our study of plug configurations explores the background pressure compensation of micro-nozzles.

2. Numerical methods and physical models

The non-equilibrium of the rarefied gas flow can be characterized by the Knudsen number ($Kn = \lambda/d$) [29], which is the ratio of the mean free path (λ) to the characteristic length (d). Different non-equilibrium regimes include slip ($0.01 < Kn \leq 0.1$), transition ($0.1 < Kn \leq 10$), and free molecular ($Kn > 10$) regimes [29]. The traditional Navier–Stokes (N-S) equations combined slip models are a common method for simulating slip flow and have been demonstrated to be accurate and efficient [30]. However, when Kn is greater than 0.1, the traditional N-S equations cannot accurately describe transition and free molecular flow [31–33], and the DSMC method [34–36] is then used. In this study, we used the CFD method combined slip model to simulate continuum and slip flows as suggested in our previous study [37]. Additionally, we chose the DSMC method for simulating transition flow in micro-nozzles because of its accurate results and computational efficiency. The methods used for various simulation cases of this study are introduced in Appendix A.

The CFD numerical simulation uses compressible N-S equations based on the assumption of laminar flow. The governing conservation equations for mass, momentum, and energy are as follows,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0, \quad (1)$$

$$\frac{\partial}{\partial t}(\rho \mathbf{V}) + \nabla \cdot (\rho \mathbf{V} \mathbf{V}) = -\nabla p + \nabla \cdot (\boldsymbol{\tau}), \quad (2)$$

$$\frac{\partial}{\partial t}(\rho E) + \nabla \cdot (\mathbf{V}(\rho E + p)) = \nabla \cdot (k \nabla T + (\boldsymbol{\tau} \cdot \mathbf{V})), \quad (3)$$

$$E = h - \frac{p}{\rho} + \frac{\mathbf{V}^2}{2}, \quad (4)$$

$$\boldsymbol{\tau} = \mu \left((\nabla \mathbf{V} + \nabla \mathbf{V}^T) - \frac{2}{3} \nabla \cdot \mathbf{V} \mathbf{I} \right), \quad (5)$$

where E is the specific energy, $\mathbf{V}$ is the velocity, ρ is the density, p is the absolute local pressure, μ is the fluid viscosity, k is the thermal conductivity, T is the static temperature, h is the enthalpy, and $\boldsymbol{\tau}$ is the viscous stress tensor. The system is closed by the ideal gas law equation of state,

$$p = \rho RT. \quad (6)$$

where R refers to the specific gas constant.

When the flow in micro-nozzles is a slip flow ($0.01 < Kn < 0.1$), the N-S equation algorithm based on continuum assumptions with a slip model is used to accurately simulate the flow inside the micro-nozzle. The first-order velocity slip u_s and temperature jump T_s based on Maxwell's theory can be expressed as follows [38,39],

$$u_s = C_1 \lambda \frac{\partial V_x}{\partial n} + V_w, \quad C_1 = \alpha \frac{2 - \sigma_v}{\sigma_v}, \quad (7)$$

$$T_s = \frac{2 - \sigma_T}{\sigma_T} \frac{2\gamma}{(\gamma + 1) Pr} \lambda \frac{\partial T}{\partial n} + T_w, \quad (8)$$

where u is the velocity. C_1 is the slip coefficient, for the first-order Maxwell slip model, the constant coefficient α is set to 1, $\sigma_v = 1$ and $\sigma_T = 1$. The subscript w represents the wall, x refers to the direction tangent to the wall, and n points towards the wall normal. The mean free path (λ) is defined as [40]

$$\lambda = A \frac{\mu}{\rho} \sqrt{\frac{1}{2\pi RT}}, \quad (9)$$

where A is a constant determined by the selected collision model in the DSMC method, and $A = 16/5$. The temperature jump model introduces the Prandtl number ($Pr = C_p \mu / \kappa$, where C_p is the specific heat at constant pressure and κ is the heat transfer coefficient) and the gas-specific heat ratio γ . When the propellant is hydrogen peroxide (H_2O_2), the calculated Pr is 0.7242, and γ is approximately 1.35. It is assumed that there is a complete energy exchange between the gas and the solid.

DSMC method is a probability-based particle simulation method proposed by Bird [34]. DSMC is a statistically-based numerical simulation method that replaces real particles in the flow field with simulated particles and decouples the collision and motion processes of simulated particles. In this study, the Larsen-Borgnakke model is used to reassign the total energy between the translational and internal modes [41]. The variable hard sphere (VHS) model [34] and the diffuse reflection model [38] are employed in our calculations. In the VHS model, the particle collision and the relative velocity between particles have a power exponential relationship. The DSMC simulation is implemented using dsmcFOAM, an OpenFOAM solver suitable for simulating rarefied gas flows in arbitrary geometries [42].

Based on a linear micro-nozzle developed by NASA GSFC [Fig. 1(a)] [1], a variety of micro-nozzles are designed to investigate the influences of geometrical configurations in this study. The propellant used is H_2O_2 . The micro-nozzle inlet gas temperature is assumed as 886 K, which is equal to the fully decomposed adiabatic flame temperature of 85% pure decomposed H_2O_2 . The micro-nozzle shown in Fig. 1(a) is employed to validate the DSMC method utilized in the study. Louissos and Hitt [12] claimed that the accuracy of a 2-D simulation for a linear micro-nozzle is close to that of a three-dimensional (3-D) simulation when the micro-nozzle is deep enough (i.e., the depth is much greater than the throat size). Under our circumstances, the depth (400 μm) is an order of magnitude bigger than the throat diameter. As a result, the 2-D simulation is employed in the following. The computational domain is configured to project 1000 μm axially and 800 μm radially from the micro-nozzle outlet, as shown in Fig. 1(b). The computational domain is similar to practical scenarios in which the surrounding micro-nozzle area is typically a solid wall. The solid walls are assumed isothermal [43], and their temperature is room temperature (300 K). The exit condition of the calculation domain is set to pressure outlet.

The grid, particle number, and time step dependence of the DSMC method are assessed. The operating pressure of the micro-nozzle is set to 10 kPa (the corresponding Re is 32), and the background pressure is 1 Pa. Three different sets of grids are chosen to test grid independence, with respective grid numbers of approximately 6000, 12000, and 30000 for coarse, standard, and refined grids. Refinement of the grids on the wall and nozzle throat involves adjusting the height of the first layer of grids at these positions while maintaining a consistent rate of

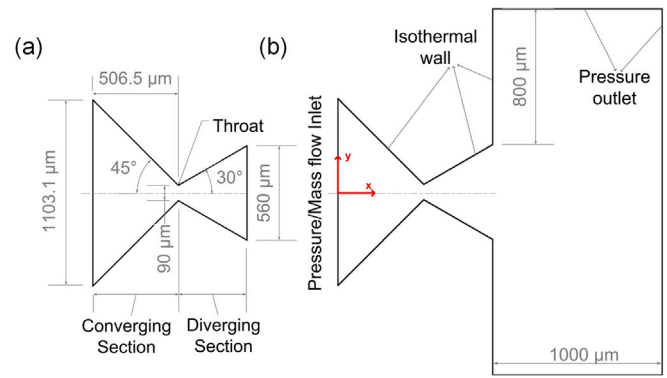


Fig. 1. (a) Schematic of the linear micro-nozzle developed by NASA GSFC [1], and the depth of the micro-nozzle is 400 μm . (b) Schematic of the computational domain.

grid size change and ensuring a desirable aspect ratio. As shown in Fig. 2(a), the simulated velocity along the micro-nozzle outlet suggests that the deviation (right Y-axis) between the moderate and refined meshes is approximately 1%, thereby demonstrating that the moderate mesh ensures sufficient simulation precision. Fig. 2(b) presents the simulation results with different particle numbers. The moderate particle count is determined to be an average of 40 colliding particles per grid cell, amounting to approximately 0.54 million particles. The coarse and refined particle counts are set at 0.27 million and 1.08 million, respectively. Based on the findings shown in Fig. 2(b), the maximum deviation between the moderate and refined particles is approximately 0.5%, and the standard particle count is then chosen. The simulation results under different time steps are presented in Fig. 2(c). The moderate time step is determined to be the mean molecular collision time, approximately $1e-9$ s. Coarse and refined time steps are set to $5e-9$ s and $5e-10$ s, respectively. Considering simulation accuracy and computational efficiency, the moderate time step is utilized for this study.

Grid independence is crucial for the CFD method. To maintain consistency with DSMC simulations, a working condition of 10 kPa and the same number of grids as in the DSMC method are employed for the CFD method. Specifically, the coarse, moderate, and refined grids are set to approximately 6000, 12000, and 30000 cells, respectively. The CFD results of the outlet velocities of the micro-nozzle under these three grid conditions are shown in Fig. 3(a). Meanwhile, due to the implementation of the slip boundary model, the simulated velocities at the micro-nozzle wall using the CFD method are analyzed, as shown in Fig. 3(b). According to the results in Fig. 3(a) and 3(b), the deviation of the results calculated by the refined mesh and the moderate mesh can essentially be ignored, and the moderate grid is selected for the CFD method in this study.

Based on the aforementioned validation results, simulations were performed for the micro-nozzle shown in Fig. 1 at the operating pressures of 1 kPa, 10 kPa, and 50 kPa, while maintaining a background pressure of 1 Pa. Fig. 4 shows the simulated results of the pressure (P), velocity (V), and Mach number (Ma). The results shown in Fig. 4(a) indicate that the micro-nozzle is under-expanded under all three operating conditions. Owing to the small scale, a thin non-equilibrium phenomenon occurs in the nozzle, leading to the occurrence of a slip velocity on its inner wall. It is observed that the slip velocity diminishes progressively as the operating pressure increases, as illustrated in Fig. 4(c). In addition, as the operating pressure decreases, the proportion of the subsonic layer area in the nozzle to the total area increases.

3. Results and discussion

In this section, the influences of the expansion ratio, cross-profile shape and plug on micro-nozzle performance are presented and discussed. Micro-nozzle performance is primarily characterized by thrust

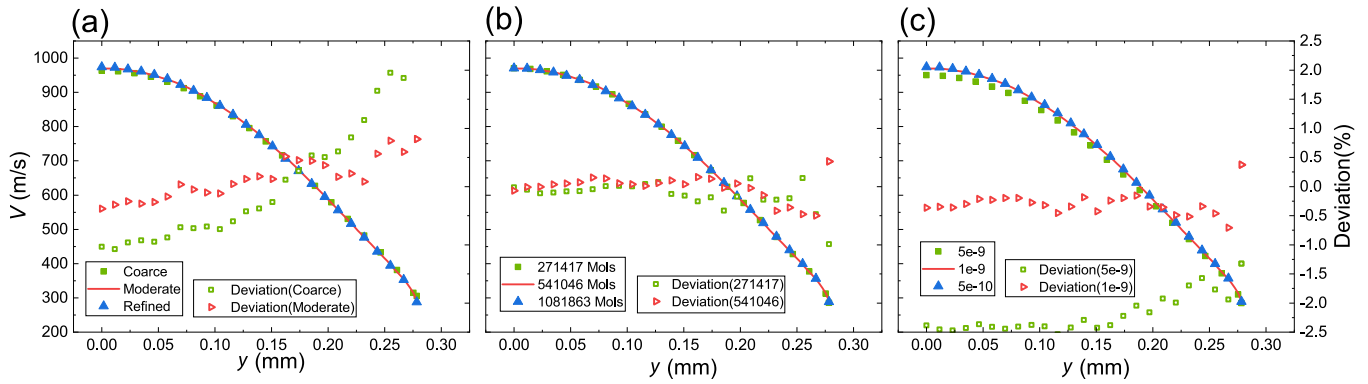


Fig. 2. DSMC simulation results of the velocity at the outlet of micro-nozzles under various (a) grid numbers, (b) particle numbers, and (c) time steps. The deviations refer to the results obtained with the coarse and standard parameters (mesh, particle, and time step) that deviate from those with the refined parameters.

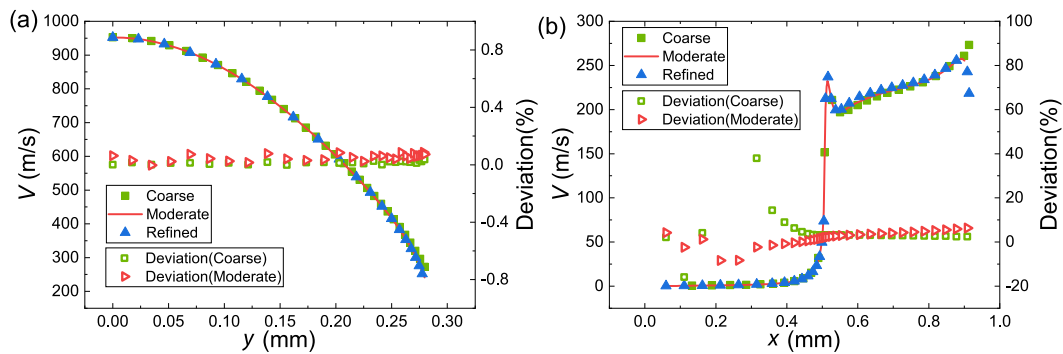


Fig. 3. CFD simulation results of the velocity at (a) the outlet and (b) the wall of micro-nozzles with various grid numbers. The deviations refer to the results obtained with the coarse and standard meshes that deviate from those with the refined meshes.

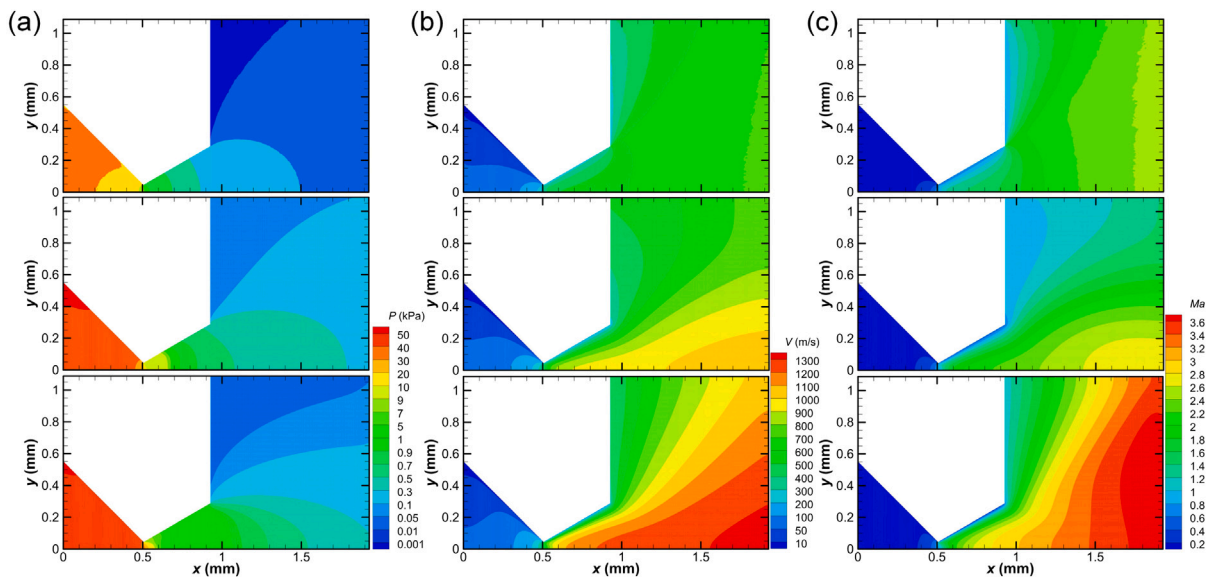


Fig. 4. The simulation results of (a) pressure (P), (b) velocity (V), and (c) Mach number (Ma) with different operating pressures. The operating pressure is 1 kPa, 10 kPa, and 50 kPa from the top row to the bottom row in sequence.

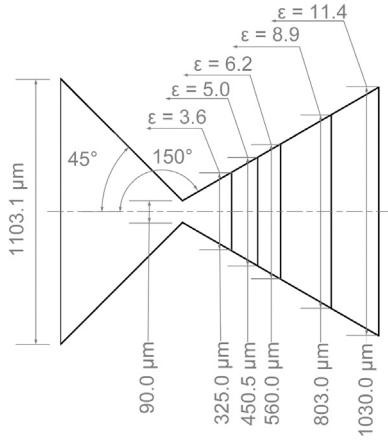


Fig. 5. Schematic of linear micro-nozzles with the different expansion ratio ϵ , which is adjusted by altering the diverging section length.

and specific impulse. The traditional thrust can be calculated as follows [37]

$$F = \int_{A_e} \rho \mathbf{V} \cdot \mathbf{n} dA_e + \int_{A_e} (P_{\text{exit}} - P_b) \mathbf{n} dA_e, \quad (10)$$

where A_e , $\mathbf{V}$, $\mathbf{n}$, and P_b are the micro-nozzle the outlet cross-profile area, the body flow velocity, the outlet section outer normal direction, and the ambient pressure, respectively; P_{exit} is the pressure at the exit of the micro-nozzle ($x = 0.92$ mm) and is calculated from the simulation. As discussed above, the surrounding area of micro-nozzles is generally a solid wall (Fig. 1). Consequently, the pressure applied to these walls diminishes the thrust generated by the micro-nozzles. The thrust of micro-nozzles can be accurately calculated from

$$F = \int_{A_e} \rho \mathbf{V} \cdot \mathbf{n} dA_e + \int_{A_e} (P_{\text{exit}} - P_b) \mathbf{n} dA_e - \int_{A_w} P_w \mathbf{n} dA_w, \quad (11)$$

where A_w is the surrounding area of the micro-nozzle outlet, and P_w is the pressure applied to the surrounding wall.

The specific impulse is calculated by [37]

$$I_{sp} = \frac{F}{\dot{m}g}, \quad (12)$$

where $\dot{m}$ is the mass flow rate of the expended propellant, g is the standard gravity. The unit of specific impulse is s in this study.

3.1. Effects of expansion ratio on the linear micro-nozzle performance

This subsection examines the effect of the expansion ratio (ϵ , defined as the ratio of the micro-nozzle exit area to the throat area) on a linear micro-nozzle. In addition to an expansion ratio of 6.2 for the original micro-nozzle (Fig. 1), four other expansion ratios of 3.6, 5.0, 8.9, and 11.4 were tested by altering the length of the diverging section (Fig. 5). Similarly, the computational domain is configured with an axial extension of 1000 μm and a radial extension of 800 μm from the micro-nozzle outlet. Taking a micro-nozzle with an expansion ratio of 3.6 as an example, a schematic of the computational domain is shown in Appendix B (see Fig. 17). Table 1 summarizes the operating conditions. Three operating pressures (P_0) were used with a background pressure of $P_b = 1$ Pa, which resulted in a throat Re ranging from 4 to 130.

Fig. 6 illustrates the variations in thrust and specific impulse with the expansion ratio at different operating pressures. At a background pressure of 1 Pa, the maximum contribution to the thrust from the background pressure acting on the wall surrounding the nozzle outlet (i.e., the third term on the right of Eq. (11)) is approximately 1.5%

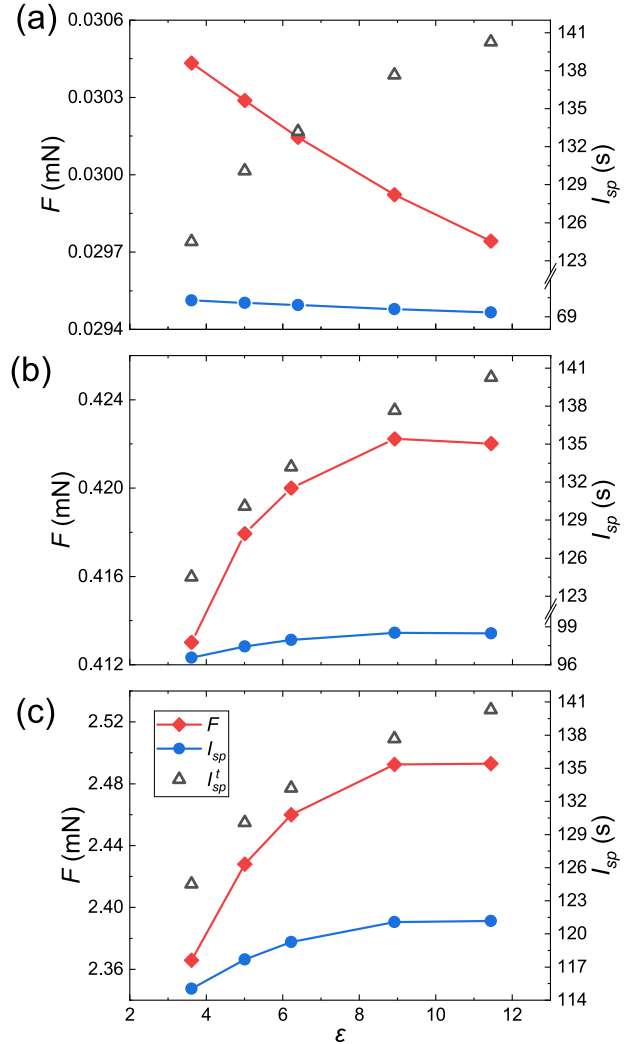


Fig. 6. Variations of micro-nozzle thrust and specific impulse with the expansion ratio (ϵ) at the operating pressures of (a) $P_0 = 1$ kPa, (b) $P_0 = 10$ kPa, and (c) $P_0 = 50$ kPa for the linear micro-nozzles shown in Fig. 5. The open symbols refer to the ideal specific impulse.

Table 1

Simulation cases for investigating the expansion ratio effect with the different operating pressures (P_0), background pressures (P_b), and Reynolds number (Re).

Cases ^a	Operating pressure (P_0)	Background pressure (P_b)	Reynolds number (Re)
Case1-1	1 kPa	1 Pa	4
Case1-2	10 kPa	1 Pa	32
Case1-3	50 kPa	1 Pa	130

^a The corresponding simulation methods are detailed in the Appendix A.

when the operating pressure is 1 kPa, and only 0.5% at the operating pressure of 50 kPa. Given its minimal contribution, this effect is not taken into account when the background pressure is 1 Pa. Consequently, Eq. (10) is employed to determine the thrust as the background pressure is 1 Pa. The results show that thrust and specific impulse decrease with an increase in the expansion ratio (or diverging section length) at $P_0 = 1$ kPa. In contrast, at $P_0 = 10$ kPa and $P_0 = 50$ kPa, both thrust and specific impulse display a positive correlation with the expansion ratio, increasing with the increase of the expansion ratio, and eventually stabilizing as $\epsilon \geq 8.9$. The performance of micro-nozzles is

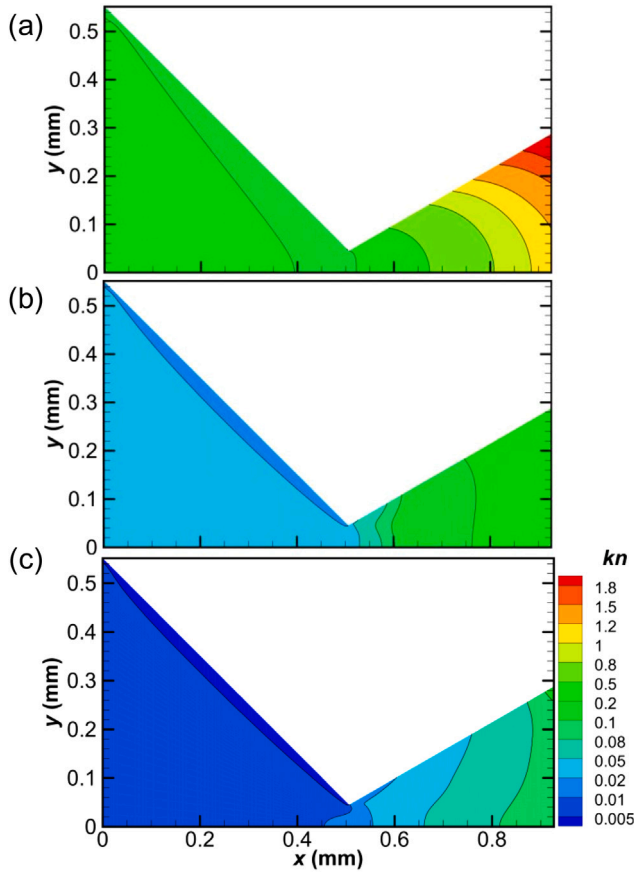


Fig. 7. Simulation results of Kn at the operating pressures of (a) $P_0 = 1$ kPa, (b) $P_0 = 10$ kPa, and (c) $P_0 = 50$ kPa for the linear micro-nozzle with the expansion ratio $\varepsilon = 6.2$.

determined by the interplay between flow expansion (expansion ratio) and friction from the subsonic layer near the micro-nozzle wall. The flow expansion improves performance, but the friction reduces it.

Fig. 7 shows the Kn number, which is calculated by dividing the local mean free path (Eq.(9)) by the characteristic geometric length d , where d is the throat diameter ($90 \mu\text{m}$) of the micro-nozzle. The results demonstrate that the flow inside the micro-nozzle can be classified as transition, slip, and continuum flow at operating pressures of $P_0 = 1$ kPa, $P_0 = 10$ kPa, and $P_0 = 50$ kPa, respectively. Fig. 8 illustrates the simulated subsonic layer near the diverging section wall for micro-nozzles with an expansion ratio of 3.6 at operating pressures of 1 kPa and 50 kPa. A thicker subsonic layer is observed at $P_0 = 1$ kPa, while a thinner layer is observed at $P_0 = 50$ kPa. The average thickness of the subsonic layer was estimated by assuming the subsonic layer as a rectangle with a length of L and a width of H , the area of which is the same as that of the actual subsonic layer, as illustrated in the insets of Fig. 8. The actual subsonic layer area is first extracted, and the width of the rectangle is then deduced by dividing this actual area by the actual subsonic length (L). The estimated rectangular widths are $H_1 = 23 \mu\text{m}$ and $H_2 = 10 \mu\text{m}$ at $P_0 = 1$ kPa and $P_0 = 50$ kPa, respectively. As the expansion ratio (ε) increases, both the exit velocity (enhancing performance) and the viscous loss caused by the subsonic layer (weakening performance) rise. At $P_0 = 1$ kPa, the viscous loss due to the subsonic layer is the predominant factor affecting the performance of the micro-nozzle, leading to a decreased performance (thrust and specific impulse) with an increase in ε [Fig. 6(a)]. At $P_0 = 50$ kPa, the influence of subsonic layer thickness still exists but is less pronounced compared to that at $P_0 = 1$ kPa. As a result, the

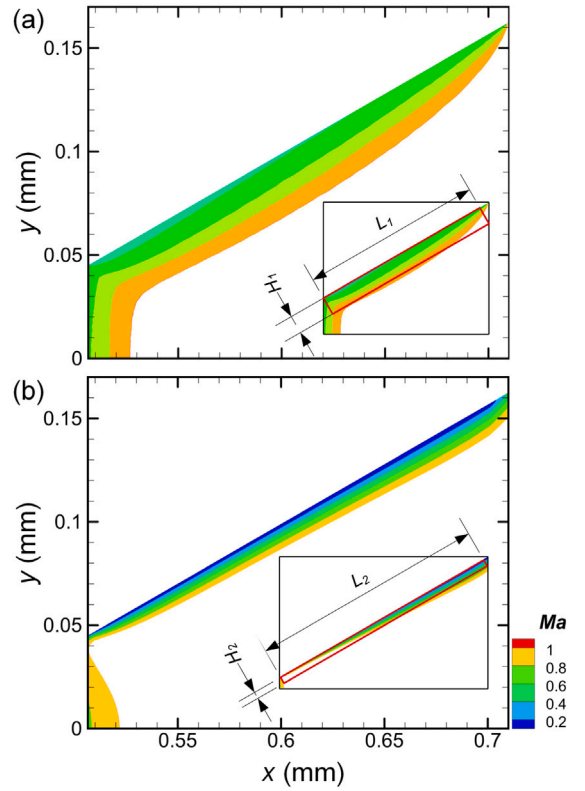


Fig. 8. Simulation results of the subsonic layer next to the diverging section wall at the operating pressures of (a) $P_0 = 1$ kPa, (b) $P_0 = 50$ kPa for the linear micro-nozzle with an expansion ratio of 3.6. The insets are the schematics of the equivalent rectangles, whose areas are the same as those of the actual subsonic layers.

performance of the micro-nozzle improves as ε increases [Fig. 6(c)]. At $P_0 = 10$ kPa, the performance is affected by both negative friction and positive expansion ratio, resulting in a modest change in performance compared to that at $P_0 = 50$ kPa. As ε varies from 3.6 to 11.4, the variation in the thrust of the linear micro-nozzle at $P_0 = 1$ kPa, $P_0 = 10$ kPa, and $P_0 = 50$ kPa is -2%, 2%, and 5%, respectively.

The nozzle specific impulse efficiency (η), defined as the ratio of the real specific impulse (I_{sp}) to the quasi-one-dimensional (ideal) specific impulse (I_{sp}^t) for a given pressure ratio, is an effective parameter for comparing the different geometries and operating conditions [12], which is expressed as

$$\eta = \frac{I_{sp}}{I_{sp}^t} = \frac{F/\dot{m}g}{\sqrt{2\gamma RT_0/[g^2(\gamma-1)]} \left[1 - (P_e/P_0)^{(\gamma-1)/\gamma}\right]}, \quad (13)$$

where P_e refers to the micro-nozzle exit pressure, which is calculated from quasi-one-dimensional theory

$$\frac{A_e}{A_t} = \frac{\sqrt{\gamma} [2/(\gamma+1)]^{(\gamma+1)/2(\gamma-1)}}{(P_e/P_0)^{1/\gamma} \sqrt{2\gamma/(\gamma-1)} \left[1 - (P_e/P_0)^{(\gamma-1)/\gamma}\right]}, \quad (14)$$

where A_e and A_t are the area of the micro-nozzle exit and throat, respectively. T_0 is the inlet gas temperature, and γ is the ratio of specific heats. The data for ideal specific impulse (I_{sp}^t) presented in Fig. 6 suggest that I_{sp}^t rises with an increase in the expansion ratio under all the operating pressure explored. As elaborated in Fig. 4(a), the micro-nozzle operates in an under-expanded state across all three examined operating conditions. By raising the expansion ratio, the gas velocity at the exit plane of the micro-nozzle is increased, which consequently increases the ideal specific impulse, as revealed by Eq. (10).

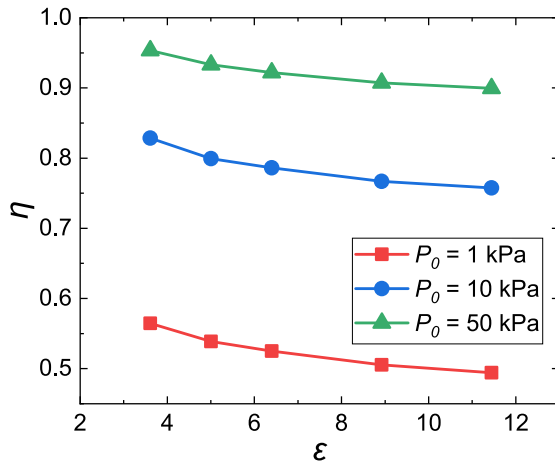


Fig. 9. The micro-nozzle specific impulse efficiency (η) at the different expansion ratios (ϵ) and operating pressures (P_0).

Fig. 9 shows the specific impulse efficiency (η) under various conditions explored, which is determined by Eq. (14). The η reduces 12.5%, 8.6%, and 5.6% from an expansion ratio of $\epsilon = 11.4$ to $\epsilon = 3.6$ at $P_0 = 1$ kPa, $P_0 = 10$ kPa, and $P_0 = 50$ kPa, respectively, suggesting that the greater the expansion ratio, the smaller the η . Taking into account the fact that the influence of friction induced by the subsonic layer is more pronounced at low operating pressure (P_0) [see Fig. 6(a)], η decreases with increasing the expansion ratio as expected when the $P_0 = 1$ kPa. At $P_0 = 10$ kPa and $P_0 = 50$ kPa, the specific impulse increases with the expansion ratio (Fig. 6). However, this increase in specific impulse comes with a trade-off: the total viscous loss also rises because of the enlarged area of the diverging section, which, ultimately, leads to a decrease in the specific impulse efficiency. The overall volume of the micro-nozzle with $\epsilon = 11.4$ is more than twice that of the micro-nozzle with $\epsilon = 3.6$. Given the preference for compact nozzle structures in practice, we highly recommend low expansion ratio micro-nozzles for micro-/nano satellites or MEMS systems.

3.2. Effects of cross profile shape on the micro-nozzle performance

This subsection examines the effect of the cross-sectional shape on the micro-nozzle performance. Linear (Fig. 1) and axisymmetric micro-nozzles with rectangular and circular cross-sections are used. Fig. 10 shows the schematic and 2-D geometries of the axisymmetric micro-nozzle. To isolate the effect of cross-sectional shape, both micro-nozzles were designed with identical inlet, throat, and exit areas, as well as converging and diverging section half-angles. As a result, the expansion ratio and mass flow rate remained the same for both micro-nozzles in this subsection. The inlet, exit, and throat diameters of the axisymmetric micro-nozzle are estimated to be 749.54 μm , 214.10 μm , and 534.05 μm for, respectively, as shown in Fig. 10. The computational domain is an analogy to that presented in Fig. 1. A schematic of the computational domain is shown in Fig. 18 of Appendix B.

The thrust and specific impulse of the linear and axisymmetric micro-nozzles were simulated for operating pressures ranging from $0.5 \leq P_0 \leq 100$ kPa using the DSMC method for $P_0 < 10$ kPa and the CFD method for $P_0 \geq 10$ kPa. The above operating pressure range corresponds to a Re spanning from 1 to 250 for the linear micro-nozzle. Since the characteristic dimension of the throat in a circular cross-section nozzle is approximately double that of a linear micro-nozzle, the Re for an axisymmetric micro-nozzle is estimated to be twice as much as that for a linear micro-nozzle. Fig. 11 shows the simulated thrust and specific impulse for both micro-nozzles at various P_0 . The average deviation in thrust between the linear and axisymmetric micro-nozzle

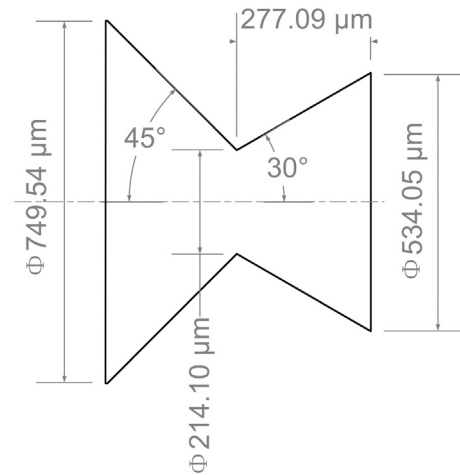


Fig. 10. Schematic of the axisymmetric micro-nozzle. The cross-sectional shape of the axisymmetric micro-nozzle is a circle, differing from the rectangle of the linear micro-nozzle shown in Fig. 1.

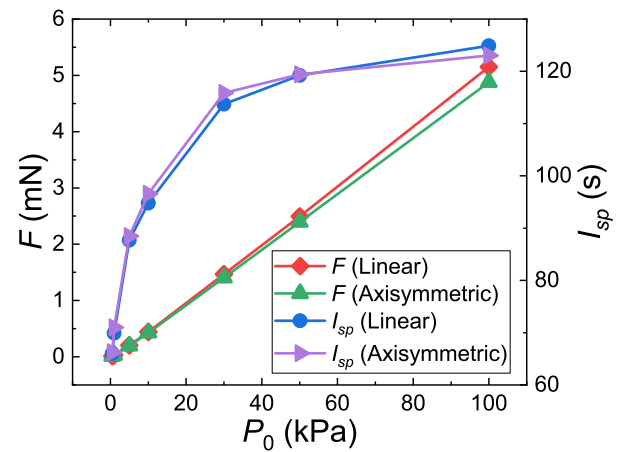


Fig. 11. Simulated thrust (F) and specific impulse (I_{sp}) of the linear and axisymmetric micro-nozzles at different operating pressures (P_0). Here the expansion ratio and propellant mass flow rate of the two micro-nozzles are the same.

is 5% within the explored P_0 range, while the deviation in specific impulse is 1%.

However, Fig. 11 shows that the thrust of the linear micro-nozzle is slightly greater than that of the axisymmetric micro-nozzle, which is unexpected. The surface area of the diverging section of the linear micro-nozzle is larger than that of the axisymmetric micro-nozzle, and hence, its thrust was expected to be smaller due to greater viscous loss. Most literature [36,44,45] uses 2-D simulation, considered an enabling approximation compared to 3-D simulation [46], to investigate micro-nozzle performance due to its high computational efficiency. Louissos and Hitt [11] noted that a deep enough micro-nozzle can minimize viscous loss for linear micro-nozzles and improve the 2-D approximation. However, the authors [12] later reported the deviation between 2-D and 3-D results, stating that 2-D simulation cannot fully account for viscous loss from the subsonic layer. In our study, the depth of the linear micro-nozzles shown in Fig. 1, Figs. 5 and 12 is 400 μm , considered sufficient for 2-D simulation [47]. However, in Fig. 10, the axisymmetric micro-nozzle has a significantly reduced depth-to-throat size ratio, implying a lack of depth. Therefore, we hypothesize that the 2-D simulation results in an unforeseen trend.

We conducted 3-D simulations, and Table 2 shows the thrust and specific impulse of the linear and axisymmetric micro-nozzles at different operating pressures. The 3-D grid constructions and representative

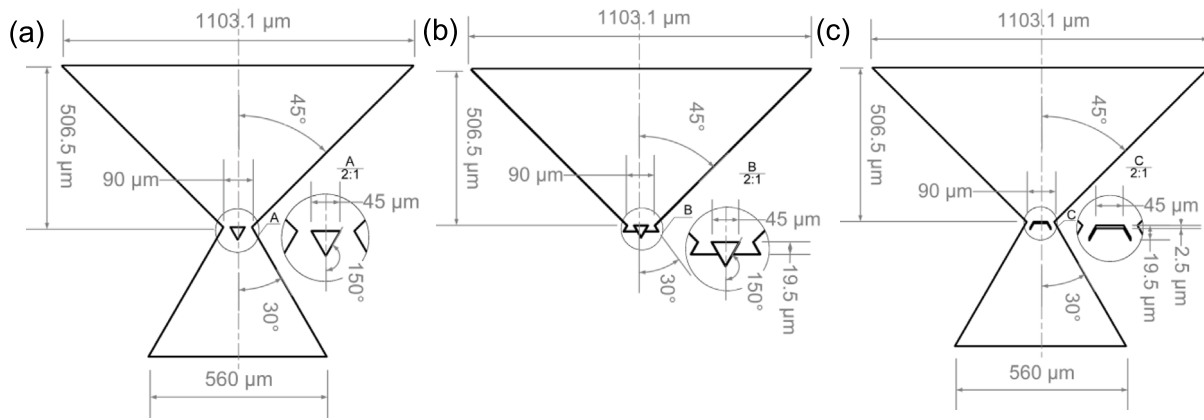


Fig. 12. Schematic of plug-nozzles with the different plug structures. They are termed as (a) S2, (b) S3, and (c) S4.

Table 2

The thrust and specific impulse of the linear and axisymmetric micro-nozzles at different operating pressures, which are obtained using the 3-D simulations.

Micro-nozzles	Operating pressure (P_0)	Thrust (F)	Specific impulse (I_{sp})
Linear	10 kPa	0.42 mN	98.00 s
Axisymmetric	10 kPa	0.46 mN	111.12 s
Linear	30 kPa	1.38 mN	107.54 s
Axisymmetric	30 kPa	1.43 mN	118.07 s
Linear	50 kPa	2.46 mN	119.27 s
Axisymmetric	50 kPa	2.50 mN	119.41 s

numerical results are presented in Appendix C. At $P_0 = 10$ kPa and $P_0 = 30$ kPa (slip flow regime), the performance of the axisymmetric micro-nozzle is better than that of the linear micro-nozzle, as expected, while their performance at $P_0 = 50$ kPa is approximately the same. The total volume of the linear (Fig. 1) and axisymmetric (Fig. 10) micro-nozzles is 0.174 mm^3 and 0.086 mm^3 , respectively, indicating that the linear micro-nozzle is twice as large as the axisymmetric micro-nozzle. Consequently, we recommend using the axisymmetric micro-nozzle instead of the linear micro-nozzle.

Our results, along with those of Ref. [12], indicate that deviations can arise in 3-D simulations. However, it remains unclear and beyond the scope of this study to determine the specific geometrical and operating conditions in which 2-D simulation can be applied. We aspire that our findings will attract future research in this field.

3.3. Effects of plugs on the micro-nozzle performance

In this subsection, we investigate micro-nozzle performance by adding plugs to the linear micro-nozzle. Four types of plug nozzles are used: structure 1 (S1, the original linear nozzle shown in Fig. 1), structure 2 (S2), structure 3 (S3), and structure 4 (S4). Fig. 12 shows the 2-D geometrical schematics of S2–S4. S2 is created by adding a triangular plug to the transitional linear micro-nozzle with an expansion ratio of 6.3 (Fig. 1). S3 is similar to S2, but its expansion ratio is reduced from 6.3 (of S2) to 1.5. S4 has the same expansion ratio as S2, but the triangular plug is replaced with a trapezoidal lamina plug. In this section, the computational domain is consistent with prior sections. As an example, a schematic of the computational domain for micro-nozzle S3 is shown in Appendix B (Fig. 19).

First, we discuss the effect of plug structures on micro-nozzle performance. In the previous section, we examined linear micro-nozzle performance at various expansion ratios. Since the inlet and throat areas remain constant, the propellant mass flow flux ($\dot{m}$) is not significantly affected by the expansion ratio at a given operating pressure. However, the throat area is dramatically altered by the plug structures

Table 3

Simulation cases for studying the effect of plug structures on the micro-nozzle at the different propellant mass flow flux ($\dot{m}$), background pressure (P_b), and Reynolds number (Re).

Cases ^a	Propellant mass flow flux ($\dot{m}$)	background pressure (P_b)	Reynolds number (Re)
Case 2–1	0.45 mg/s	1 Pa	32
Case 2–2	2.14 mg/s	1 Pa	130

^a The corresponding simulation methods are detailed in the Appendix A.

shown in Fig. 12. Therefore, we used a constant $\dot{m}$ to investigate the influence of plug structures on micro-nozzle performance, as shown in Case 2–1 and Case 2–2 of Table 3. The inlet pressures of the nozzles S1, S2, S3, and S4 at a propellant mass flow of $\dot{m} = 0.45 \text{ mg/s}$ are $P_0 = 10 \text{ kPa}$, $P_0 = 20 \text{ kPa}$, $P_0 = 20 \text{ kPa}$, and $P_0 = 27 \text{ kPa}$, respectively. At $\dot{m} = 2.14 \text{ mg/s}$, the corresponding inlet pressures for the micro-nozzle-S1, S2, S3, and S4 are $P_0 = 50 \text{ kPa}$, $P_0 = 100 \text{ kPa}$, $P_0 = 100 \text{ kPa}$, and $P_0 = 127 \text{ kPa}$, respectively. It should be noted that the background pressures for Case 2–1 and Case 2–2 are identical at $P_b = 1 \text{ Pa}$.

Fig. 13 shows simulation results of thrust and specific impulse with propellant mass flow flux of $\dot{m} = 0.45 \text{ mg/s}$ (Case 2–1) and $\dot{m} = 2.14 \text{ mg/s}$ (Case 2–2). The results indicate that, compared to the original linear micro-nozzle (i.e., S1), the plug structures explored in our study significantly reduce micro-nozzle performance. This is primarily due to the dramatic reduction in gas velocity at the exit plane caused by the plug structures, which degrades micro-nozzle performance. Fig. 14 displays the simulated Kn of the internal flow for the S2, S3, and S4 plug micro-nozzles at different propellant mass flow rates. The transition flow ($Kn \geq 0.1$) occurs in the diverging section of the internal flow at a propellant mass flow rate of $\dot{m} = 0.45 \text{ mg/s}$ (Case 2–1), while the slip flow ($0.01 \leq Kn < 0.1$) occurs at a propellant mass flow rate of $\dot{m} = 2.14 \text{ mg/s}$ (Case 2–2).

At $\dot{m} = 0.45 \text{ mg/s}$ (the transition flow regime), the S2 plug micro-nozzle performs the best among the three plug micro-nozzles, while the S4 plug micro-nozzle performs the worst, as shown in Fig. 13(a). As previously discussed, micro-nozzle performance is determined by the viscous loss and the degree of flow expansion. On the one hand, the trapezoidal lamina plug structure of the S4 micro-nozzle is parallel to the diverging section walls, inhibiting flow expansion compared to the triangle plug structure of the S2 micro-nozzle. On the other hand, viscous loss plays a crucial role in the slip flow regime. The diverging section of the S4 micro-nozzle is significantly larger than that of the S3 micro-nozzle, resulting in greater viscous loss and, consequently, the worst performance among the three micro-nozzles. At $\dot{m} = 2.14 \text{ mg/s}$ (the slip flow regime), the S2 plug micro-nozzle performs the best, while the S3 plug micro-nozzle performs the worst,

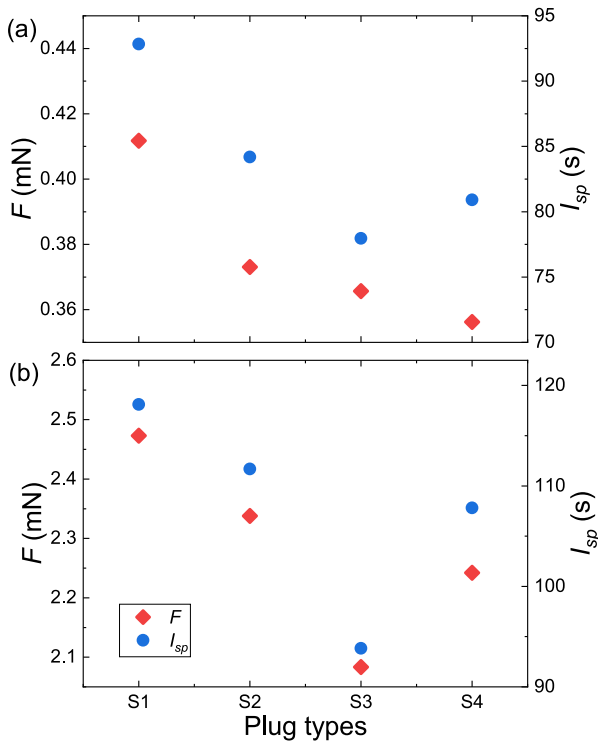


Fig. 13. The thrust and specific impulse of the micro-nozzle at the propellant mass flow flux of (a) $\dot{m} = 0.45 \text{ mg/s}$, (b) $\dot{m} = 2.14 \text{ mg/s}$.

Table 4

Simulation cases for studying the background compensation of plug micro-nozzles at different operating pressures (P_0), background pressures (P_b), and Reynolds number (Re)

Cases ^a	Operating pressure (P_0)	Background pressure (P_b)	Reynolds number (Re)
Case 3–1	10 kPa	1 Pa	32
Case 3–2	50 kPa	1 Pa	130
Case 3–3	10 kPa	2000 Pa	20
Case 3–4	50 kPa	2000 Pa	70

^a The corresponding simulation methods are detailed in the Appendix A.

as shown in Fig. 13(b). Since the expansion ratio dominates micro-nozzle performance in the early slip and continuum flow regimes, the thrust and specific impulse of the S3 micro-nozzle are the smallest.

Second, the background pressure compensation for the plug micro-nozzles is discussed. Two operating pressures ($P_0 = 10 \text{ kPa}$ and $P_0 = 50 \text{ kPa}$) and two kinds of background pressures ($P_b = 1 \text{ Pa}$ and $P_b = 2000 \text{ Pa}$) are considered, as shown in Table 4.

Fig. 15 shows the simulated thrust and specific impulse of plug micro-nozzles at different operating and background pressures. As mentioned earlier, the thrust at a background pressure of 1 Pa can be obtained using Eq. (10). Nevertheless, the background pressure acting on the surrounding wall can lead to a reduction in micro-nozzles' thrust of up to 13%, and hence, the third term on the left side of Eq. (11) cannot be ignored. Consequently, the thrust under the background pressure of 2000 Pa is assessed through Eq. (11). At an operating pressure of $P_0 = 10 \text{ kPa}$, the deviations between the thrust obtained at $P_b = 1 \text{ Pa}$ and that obtained at $P_b = 2000 \text{ Pa}$ are 62%, 89%, 51%, and 95% for the S1, S2, S3, and S4 micro-nozzles, respectively, as shown in Fig. 15(a). Similarly, at $P_0 = 50 \text{ kPa}$, the deviations between the thrust simulated at $P_b = 1 \text{ Pa}$ and that obtained at $P_b = 2000 \text{ Pa}$ are 19%, 35%, 10%, and 41% for the S1, S2, S3, and S4 micro-nozzles, respectively, as shown in Fig. 15(b). The variation of specific impulse coincides with that of the thrust. Compared to other plug micro-nozzles,

the variation of thrust and specific impulse with background pressure is the smallest for the S3 plug micro-nozzles. The S3 plug micro-nozzle has a semi-open nozzle structure that can automatically adapt to external counter-pressure changes to avoid under-expansion. The S3 type is least affected by background pressure. However, under the same conditions, the original linear micro-nozzle (S1 shown in Fig. 1) still performs best in terms of thrust and specific impulse.

Fig. 16 presents the typical Ma of the S2 and S3 plug micro-nozzles at an operating pressure of $P_0 = 10 \text{ kPa}$ and background pressure of $P_b = 1 \text{ Pa}$ and $P_b = 2000 \text{ Pa}$. The results indicate that at a background pressure of $P_b = 2000 \text{ Pa}$, ambient flow can communicate with the diverging section of the S2 plugs micro-nozzle, leading to backflow, deflection of exit momentum in the transverse direction, and degradation of performance, as shown in Fig. 16(b). On the contrary, the S3 plug micro-nozzle is under-expanded at a background pressure of $P_b = 2000 \text{ Pa}$ due to its truncated diverging section. Hence, the performance of the micro-nozzle with the S3 plug is better than that with the S2 plug. The S4 micro-nozzle, with its trapezoidal lamina plug, hinders flow expansion, reducing its performance compared to the S2 micro-nozzle.

In summary, the S3 plug micro-nozzle is suitable for a wide range of pressure environments, although it may not have the highest thrust and specific impulse. This makes it beneficial for potential applications such as heat sinks for microelectronic fabrication systems, where the external environment is highly variable. Intriguingly, the experimental results reported by Ketsdever et al. [7] indicate that the thrust of the thin-walled orifice, whose configuration is similar to that of the S3 plug micro-nozzle in our study, is superior to that of common micro-nozzles at the low Reynolds number range ($Re < 100$). It is attributed to the orifice's smaller wall area, which results in reduced viscous losses. However, the micro-nozzle still maintains a higher efficiency in terms of specific impulse compared to the orifice. Moreover, the flow field (temperature, velocity, and pressure) distributions downstream of the micro-nozzle are more uniform compared to those of the thin-walled orifice, being beneficial to the application spray cooling.

Our findings suggest that complex structural designs could result in more efficient micro-nozzle configurations. Therefore, in future micro-nozzle studies, the analysis of performance for micro-nozzle configurations with more intricate designs, which could further enhance their efficiency, is encouraged. Additionally, we note that there is still a significant gap in experimental results for micro-nozzles. As such, experimental studies are supposed to be conducted in the future to gather comprehensive data that could validate and complement our computational simulations of micro-nozzles.

4. Conclusion

In this study, the influence of the expansion ratio, cross profile shapes, and plugs on micro-nozzle performance (thrust and specific impulse) is numerically investigated using CFD and DSMC methods. The expansion ratio is varied by altering the length of the diverging section. The results show that at an operating pressure of $P_0 = 1 \text{ kPa}$ (transition regime), the thrust and specific impulse of the linear micro-nozzle decrease with increasing expansion ratio, while they increase at $P_0 = 10 \text{ kPa}$ (slip regime) and $P_0 = 50 \text{ kPa}$ (continuum regime). The former is due to the enhancement of the subsonic layer next to the micro-nozzle wall, increasing friction loss. The latter benefits from increased flow expansion. Specifically, the results reveal that the micro-nozzle efficiency decreases with increasing expansion ratio and decreasing operating pressure.

The thrust and specific impulse of linear and axisymmetric micro-nozzles with rectangular and circular cross-profile shapes are simulated. The results display that the performance of the axisymmetric micro-nozzle is better than that of the linear micro-nozzle at the operating pressure of $P_0 \leq 10 \text{ kPa}$ (rarefied region), attributed to the larger surface area of the linear micro-nozzle causing greater viscous loss. At

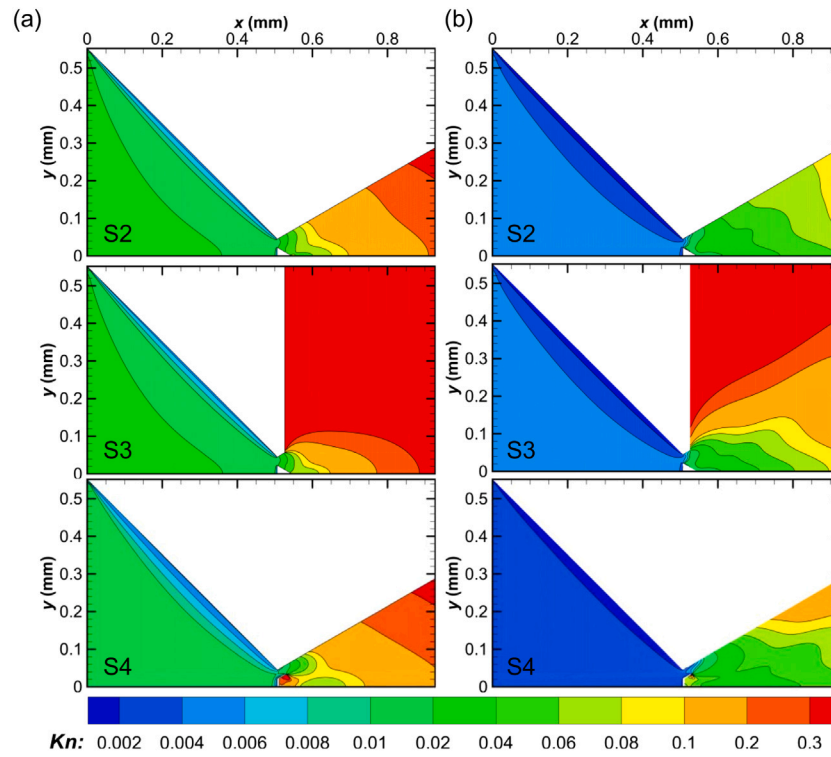


Fig. 14. Simulation results of Kn for the S2, S3, and S4 plug micro-nozzles at the propellant mass flow flux of (a) $\dot{m} = 0.45 \text{ mg/s}$ (Case2-1), (b) $\dot{m} = 2.14 \text{ mg/s}$ (Case2-2).

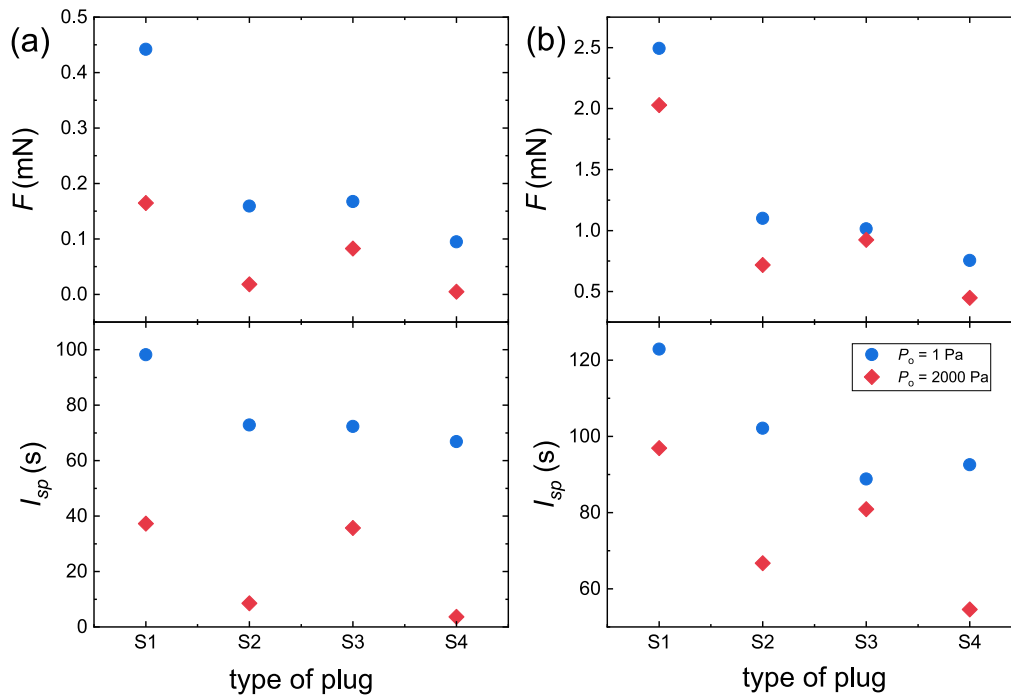


Fig. 15. The thrust and specific impulse of the plug micro-nozzles at the operating pressure of (a) $P_0 = 10 \text{ kPa}$, (b) $P_0 = 50 \text{ kPa}$.

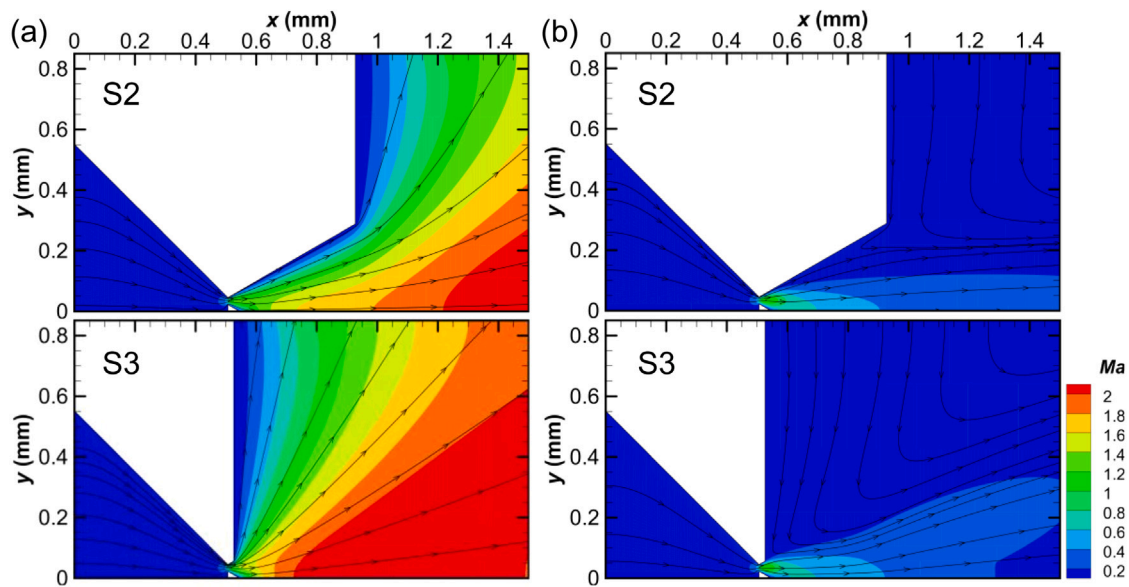


Fig. 16. Simulation results of Ma for plug micro-nozzles of S2 and S3 at the operating pressure of $P_0 = 10$ kPa and background pressure of (a) $P_b = 1$ Pa (Case3-1) as well as (b) $P_b = 2000$ Pa (Case3-3).

$P_0 = 50$ kPa (continuum regime), their performances are approximately identical, as expected, since the viscous loss has only a minor effect on micro-nozzle performance in the continuum flow regime. Nevertheless, the axisymmetric micro-nozzle is always recommended due to its comprehensive performance and size.

The background pressure compensation is examined using three types of plug micro-nozzles, constructed by adding different plug structures to the traditional linear micro-nozzle. The results indicate that the plug structures significantly reduce the thrust and specific impulse of the micro-nozzle. However, the S3 plug micro-nozzle [Fig. 12(b)], with a triangular plug and an expansion ratio of 1.5, has the advantage of being suitable for a wide range of background pressures, despite not having the highest thrust and specific impulse among the three plug micro-nozzles explored. This is beneficial for potential applications where the external environment changes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Simulation methods for different conditions

Our previous research [37] pointed out that for operating pressures of the micro-nozzle shown in Fig. 1(a) exceeding 10 kPa (i.e., for the slip flow and continuum flow), the Computational Fluid Dynamics (CFD) combined the slip model can produce results within an acceptable error. Conversely, for operating pressures below 10 kPa (i.e., for the transition flow and rarefied flow), the application of CFD yields larger errors, and it is necessary to use the direct simulation Monte Carlo (DSMC) method. The methodologies applied in our study are detailed in Table A.5. In particular, the slip model used here is the Maxwell model, which is introduced detailedly in Ref. [37].

To illustrate the discrepancy between the DSMC and CFD-slip methods, three representative cases are simulated by the DSMC and CFD-slip methods simultaneously. Table A.6 shows the simulated thrust with different simulation methods for the linear micro-nozzle shown in Fig. 1. When the operating pressure is 1 kPa, the deviation between the DSMC and CFD methods is 21%. On the contrary, with operating pressures exceeding 10 kPa, the deviation diminishes to less than 3.5%. These findings further validate the simulation methods selected in our study.

Table A.5
The simulation methods used in various conditions.

Cases	Method
Case 1-1	DSMC
Case 1-2	CFD+slip model
Case 1-3	CFD+slip model
Case 2-1	CFD+slip model
Case 2-2	CFD+slip model
Case 3-1	CFD+slip model
Case 3-2	CFD+slip model
Case 3-3	CFD+slip model
Case 3-4	CFD+slip model

Table A.6
The simulated thrust with different simulation methods for the linear micro-nozzle shown in Fig. 1.

Cases	DSMC	CFD+Slip	Deviation(%)
Case 1-1 ^a	0.031 mN	0.024 mN	21.00
Case 1-2 ^b	0.44 mN	0.42 mN	3.25
Case 1-3 ^c	2.52 mN	2.46 mN	2.50

^a The operating pressure is 1 kPa.
^b The operating pressure is 10 kPa.
^c The operating pressure is 50 kPa.

Appendix B. Schematic of the computational domain for various micro-nozzles

In this section, we present the computational domain examples for the various micro-nozzles explored in our study. Fig. 17, 18, and 19 are the computational domain for the micro-nozzle with an expansion ratio of 3.6 (Fig. 5), the axisymmetric micro-nozzle (Fig. 10), and the S3 plug micro-nozzle (Fig. 12), respectively.

Appendix C. 3-D micro-nozzle simulation

To ensure consistency in our results, we applied the same mesh setting strategy for both linear and axisymmetric micro-nozzles. To ensure the accuracy of the simulation, all mesh shapes are quadrilateral, with the smallest grid size at the wall, gradually increasing towards the axis, as shown in Fig. 20. We used the mean free path ($\lambda = 3 \mu\text{m}$) at the throat with an operating pressure of 10 kPa as the size of the first layer

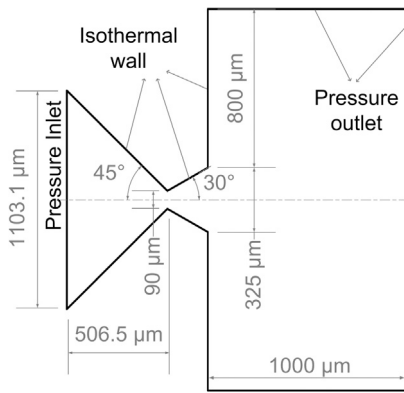


Fig. 17. Schematic of the computational domain for the micro-nozzle with an expansion ratio of 3.6.

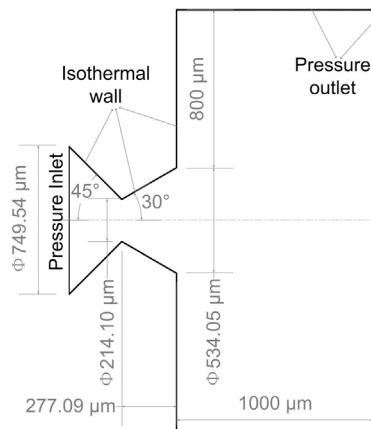


Fig. 18. Schematic of the computational domain for the axisymmetric micro-nozzle.

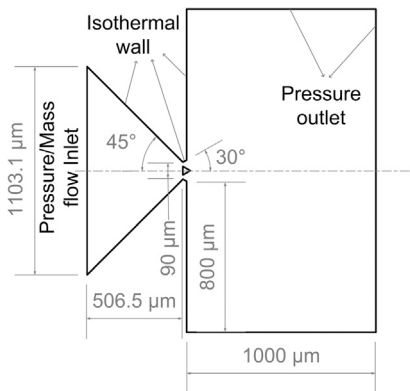


Fig. 19. Schematic of the computational domain for the S3 plug micro-nozzle shown in Fig. 12.

of mesh with a mesh expansion ratio is 1.05. We ensure that all results are insensitive to further grid refinement (see Fig. 21).

Fig. 21 shows the simulated Ma of the linear and axisymmetric micro-nozzles. The Ma result of the 3-D linear micro-nozzle reveals viscous subsonic layers extending from the expander walls, resulting in portions of the flow cross-profile becoming subsonic. This difference in Ma is the main reason for the difference in performance between the linear and axisymmetric micro-nozzles.

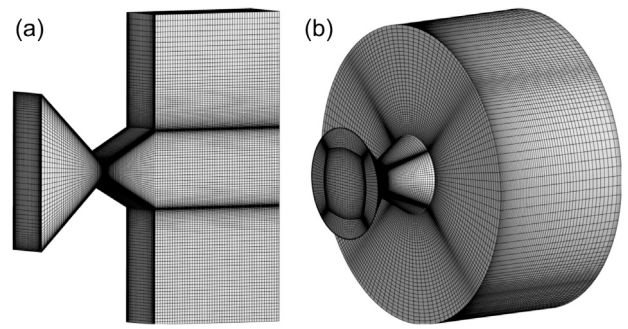


Fig. 20. The 3-D mesh of the (a) linear and (b) axisymmetric micro-nozzles.

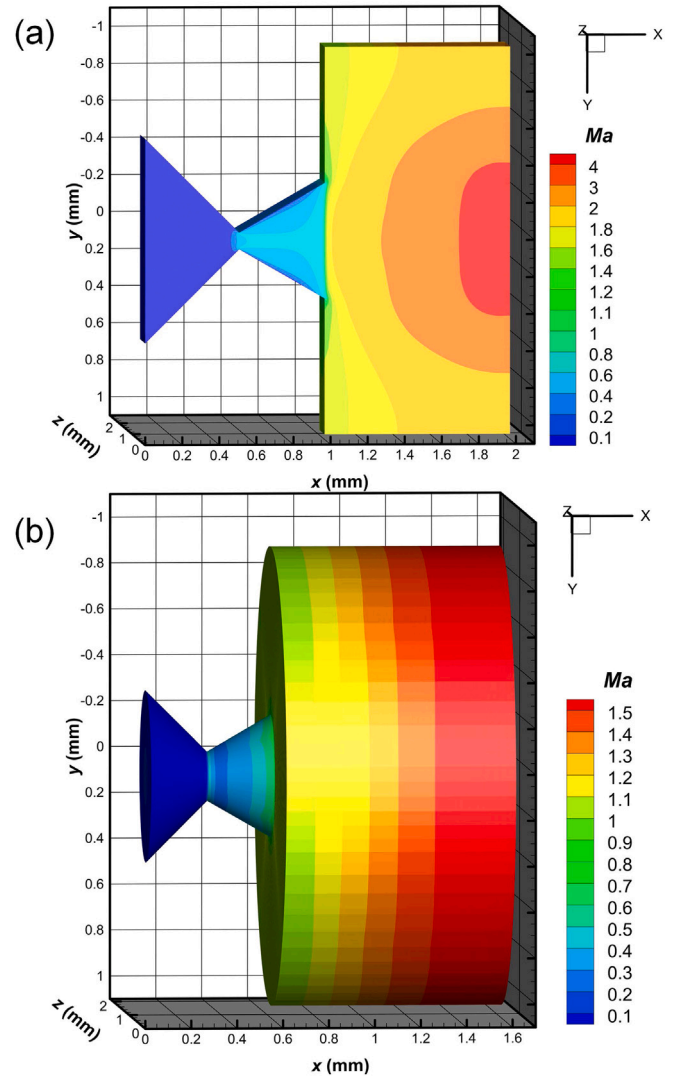


Fig. 21. The simulated Ma of the (a) linear and (b) axisymmetric micro-nozzles.

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